Lab 2: LLM- and RAG-based Multi-hop Question Answering System

SE3332 Machine Learning

Yukai Wang
Student ID: 523111910158

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Contents

1 System Implementation **2**
1.1 Overall Architecture . . . 2
1.2 Dataset and Context Structure . . . 2
1.3 Context Chunking . . . 2
1.4 BM25 Retrieval . . . 2
1.5 Generation . . . 2

2 Evaluation Results **2**
2.1 Metrics . . . 2
2.2 Results . . . 3

3 Error Analysis (Case Study) **3**
3.1 F1 Score Distribution . . . 3
3.2 Error Type 1: Vocabulary Mismatch (Primary Error, 60% of failures) . . . 3
3.3 Error Type 2: Over-specific Answers (5% of failures) . . . 4
3.4 Error Type 3: Multi-hop Chain Breaks (35% of failures) . . . 4

4 Improved Strategy: Query Decomposition **4**
4.1 Motivation . . . 4
4.2 Method . . . 4
4.3 Results and Analysis . . . 5
4.4 Why Decomposition Hurt . . . 5
4.5 Discussion and Future Directions . . . 5

5 Conclusion **6**

1 System Implementation

1.1 Overall Architecture

The system consists of two key stages: **retrieval** (finding the passages most likely to contain the answer) and **generation** (prompting a large language model to produce a concise answer from the retrieved text). Both stages are described below.

1.2 Dataset and Context Structure

The HotpotQA-LongBench dataset contains 200 multi-hop questions. Each entry includes a `context` field that contains 2–10 passages (average 8.6), separated by `Passage N:` markers. Context lengths range from 6,558 to 81,304 characters (average 56,550), making the full context too long to supply directly to a generation model without selection.

1.3 Context Chunking

We first split each context along `Passage N:` markers to isolate individual Wikipedia passages. Passages that contain fewer than 150 words are kept as single chunks. For longer passages, we apply a sliding-window strategy with a **window of 150 words** and an **overlap of 30 words** to prevent answers that straddle chunk boundaries from being split. Empirically, the 150-word window is small enough to keep chunk semantics focused while large enough to preserve sentence-level context.

1.4 BM25 Retrieval

Given a query, we use **Okapi BM25** (via the `rank-bm25` library) to score all chunks and return the top-k ($k = 5$) highest-scoring chunks. BM25 is a strong sparse retrieval baseline that requires no additional model downloads or GPU compute, while still capturing exact-match and TF-IDF-like signals.

1.5 Generation

Retrieved chunks are concatenated and passed to **GPT-4o-mini** via the OpenAI API with a carefully engineered system prompt instructing the model to:

- Answer *only* from the provided context.
- For Yes/No questions: output only "Yes" or "No".
- For all other questions: output only the exact answer phrase (name, date, place, number, or short phrase)—no full sentences.

Temperature is set to 0.1 to minimize hallucination and maximise determinism.

2 Evaluation Results

2.1 Metrics

We use two standard metrics from the SQuAD/HotpotQA literature:

- **EM (Exact Match)**: 1 if the normalised predicted answer equals the normalised gold answer, else 0.
- **F1**: token-level F1 between predicted and gold answer after normalisation (lower-case, remove punctuation and articles).

Table 1: System performance on 200-sample HotpotQA-LongBench

System	Avg EM	Avg F1
No RAG (baseline, example.py)	0.2000	0.2467
Basic RAG (BM25, ours)	**0.4150**	**0.5266**
Query-Decomp RAG (improved, ours)	0.3450	0.4492

2.2 Results

The Basic RAG system improves EM by **+107%** and F1 by **+113%** over the no-RAG baseline, demonstrating the effectiveness of retrieval augmentation. The query decomposition system is analysed in detail in Section 4.

3 Error Analysis (Case Study)

3.1 F1 Score Distribution

Table 2: F1 score distribution for the Basic RAG system

F1 range	Count	%
1.0 (perfect)	83	41.5%
0.7 – 1.0	10	5.0%
0.3 – 0.7	22	11.0%
0 – 0.3	8	4.0%
0.0 (total miss)	77	38.5%

The distribution is *bimodal*: the system either answers correctly (41.5% with F1=1.0) or completely fails (38.5% with F1=0.0). This strongly suggests that **retrieval quality** is the dominant bottleneck: when BM25 retrieves the correct passage, GPT-4o-mini almost always extracts the right answer; when it does not, the model has no useful information and fails.

3.2 Error Type 1: Vocabulary Mismatch (Primary Error, 60% of failures)

BM25 is a bag-of-words method. If the question uses different vocabulary than the answer passage, retrieval will fail regardless of answer quality.

Example:
Q: Which case was brought to court first, Miller v. California or Gates v. Collier?
Gold: *Miller v. California*
Prediction: *Gates v. Collier*

Here, the key linking fact (dates) is buried in a passage about "trusty system" and "Parchman Farm", which does not match the legal vocabulary of the question well. BM25 retrieves passages mentioning both case names but misses the critical date comparison.

Example:
Q: Prior to playing for Michigan State, Keith Nichol played football for a school located in what city?
Gold: *Norman*
Prediction: *Lowell, Michigan*

BM25 retrieves chunks mentioning "Michigan State" and "Keith Nichol" but misses the earlier passage about his high school in Norman, Oklahoma, because the question's phrasing doesn't lexically match that passage.

3.3 Error Type 2: Over-specific Answers (5% of failures)

In 10 cases the model produces an answer that contains the gold answer as a substring, but includes additional detail. For example:

- Gold: *Nobel Prize* Prediction: *Nobel Prize in Physics*
- Gold: *Manchester United* Prediction: *Manchester United F.C.*
- Gold: *November* Prediction: *November 23, 2002*

These fail EM but get high F1. The issue is that the model is "too precise" relative to what the dataset annotators wrote.

3.4 Error Type 3: Multi-hop Chain Breaks (35% of failures)

Some questions require facts from two separate passages. BM25 retrieves the top-5 chunks by a single similarity score, so it may retrieve 5 chunks all from the same passage while missing the second hop entirely. For example:

Example:
Q: Gary L. Bennett was part of space missions with what primary destination?
Gold: *Sun*
Prediction: *Jupiter, Saturn, Uranus, Neptune*

The model retrieves passages about Bennett's missions (outer planets) but misses the passage about a solar probe mission. The two relevant passages are not retrieved together.

4 Improved Strategy: Query Decomposition

4.1 Motivation

The error analysis reveals that the primary failure mode is BM25 retrieval with a single holistic query. Multi-hop questions semantically span multiple passages, and a single BM25 query may score well against only one hop. Query Decomposition (as described in the lecture and in the lab background) addresses this by breaking the question into atomic sub-questions, each targeting a single hop.

4.2 Method

The improved pipeline has three stages:

1. **Decompose**: GPT-4o-mini receives the original multi-hop question and is instructed to produce 2–3 simpler sub-questions, each independently answerable from a single passage.
2. **Iterative RAG**: Each sub-question is answered in order using BM25 retrieval. Crucially, prior sub-answers are *concatenated into the next retrieval query*, enabling cross-hop evidence chaining: the answer to hop 1 enriches the query for hop 2.
3. **Synthesis**: A final GPT call receives all sub-question/answer pairs and produces the definitive answer to the original question.

Table 3: Query-Decomp RAG vs. Basic RAG breakdown

	Basic wins	**Decomp wins**	**Both correct**	**Both wrong**
Cases	41	22	58	79

4.3 Results and Analysis

The decomposition system improves 22 cases (where direct BM25 misses but sub-question queries succeed) but *regresses* on 41 cases compared to the basic system. Overall EM drops from 0.415 to 0.345.

4.4 Why Decomposition Hurt

We identify four root causes:

1. **Over-specific synthesis**: The synthesis LLM sometimes produces more detailed answers than the gold annotation expects. For example, for the question "After Russell D. Moore became President of the ERLC, in what year?" the gold is *2013* but the decomp system outputs *June 1, 2013*.
2. **Error propagation**: If sub-question 1 is answered incorrectly, sub-question 2 uses a wrong premise, and the synthesised final answer inherits both errors. The basic RAG system makes a single retrieval and generation step with no error propagation.
3. **Abstention**: In some cases the iterative retrieval for a sub-question returns chunks with no relevant information. The model outputs "Unknown", which causes the synthesis step to fail completely. The basic RAG system, retrieving directly with the full question, is more likely to find at least a partial match.
4. **Imperfect decomposition**: GPT-4o-mini does not always decompose questions in the optimal way. Sub-questions may overlap, be too broad, or miss the key bridging entity entirely. Decomposition quality is sensitive to prompt wording and model capability.

Cases where decomposition helped (22 cases): These are genuinely multi-hop questions where the original question's vocabulary did not match any single passage, but atomic sub-questions each matched their target passage directly. For instance, "Which case was brought to court first, Miller v. California or Gates v. Collier?" was correctly answered by decomposing into "When was Miller v. California decided?" and "When was Gates v. Collier decided?".

4.5 Discussion and Future Directions

The experiment confirms that **query decomposition is not universally beneficial**: it helps when decomposition is accurate and sub-questions are well-matched to distinct passages, but it introduces new failure modes (over-specificity, error propagation, abstention) that outweigh the gains on this dataset.

Future improvements could include:

- **Dense retrieval**: Replace BM25 with sentence-transformer embeddings (e.g., `all-MiniLM-L6-v2`) to capture semantic similarity beyond exact lexical overlap. This would address vocabulary mismatch without introducing decomposition complexity.
- **Hybrid retrieval**: Combine BM25 and dense scores with linear interpolation (e.g., RRF fusion) to get the benefits of both.
- **Confidence-gated decomposition**: Only apply decomposition when the direct BM25 retrieval score is low, falling back to the basic RAG otherwise.

- **Self-consistency**: Sample multiple generations and take a majority vote to reduce variance.
- **Larger top-k + LLM re-ranking**: Retrieve $k = 20$ chunks and ask the LLM to select the 3 most relevant before answering, reducing the impact of BM25's ranking errors.

5 Conclusion

We built a RAG-based multi-hop QA system using BM25 retrieval and GPT-4o-mini generation. The basic system achieves EM=0.4150 and F1=0.5266, more than doubling the no-RAG baseline (EM=0.200, F1=0.247). Error analysis reveals a bimodal success distribution driven by retrieval quality. We implemented Query Decomposition as an improved strategy; while it helps for genuinely multi-hop questions, it introduces error propagation and over-specificity that lower overall performance. The analysis provides clear insights into the trade-offs and motivates dense retrieval as the most promising next step.

Code and External Resources: All code is original. The evaluation functions (`normalize_answer`, `compute_em`, `compute_f1`) are adapted from the provided `example.py`. The `rank-bm25` Python library is used for BM25 retrieval. The LLM API used is OpenAI GPT-4o-mini. Source code is available at https://github.com/wangyukai585/SE3332-lab2.